

BEACON DATA PROCESSING SYSTEM (BPS)

*Comprehensive System Architecture & Functional Specifications Manual
NASA/CREST Project Technical Documentation*

Document Configuration History

Author	Version Date	Change Summary
Jim Pham	May 2, 2008	Initial draft - V 0.1
Edward Anthonysamy	May 11, 2008	Draft - V 0.2
Veeraragavan Gopalakrishnan	May 12, 2008	Draft - V 0.3
Bruce Davis	May 22, 2008	Draft - V 0.4
Edward Anthonysamy	May 23, 2008	Draft - V 0.5

1. Project Overview

The Beacon Data Processing System (BPS) is an automated, multi-tier web platform engineered to manage operational satellite telemetry data. Developed in partnership with Santa Clara University for the Center for Robotics Exploration and Space Technologies (CREST) and NASA, BPS serves as a standard operational gateway for external amateur radio (HAM) satellite operators to submit real-time telemetry packets. These packet strings are received from spaceborne platforms including GeneSat-1, PreSat, and PharmaSat. The platform systematically archives, processes, and parses incoming streams to provide search, diagnostic query, and statistical reporting capabilities to researchers, educators, and public institutions.

2. System Architecture & Detailed Data Pipeline

The application architecture leverages a decoupled layout hosted on Microsoft Internet Information Services (IIS). The system routes structural satellite telemetry from presentation forms down to transactional persistence via five highly coordinated layers:

- **Presentation Tier (Beacon Client Application — ASPX):** Constructed on the Microsoft .NET WebForms framework using C# code-behind binaries. It unifies user navigation via an XML site map control mapped to a global Master Page layout. It provides validation forms

for HAM operators to key in alphanumeric callsigns, structural emails, coordinates, and multi-line hex packet entries.

- **Orchestration Tier (ASP.NET Request Handler & Web Services — ASMX):** Managed directly via the IIS worker pipeline. The incoming traffic is passed through the ASP.NET request handler into exposed SOAP-compliant web service modules. These public APIs expose standard service methods (such as ProcessBeaconData and GetPacketsbyDateRange) allowing automated telemetry ingest from external script nodes, university terminals, and space center hubs.
- **Domain Business Logic Tier (Beacon Data Format Library — .DLL):** Acts as the central programmatic calculation core. This component executes circular-endian byte mapping algorithms to extract raw hexadecimal substrings and cast them into computational integer values. During user query operations, it transparently applies calibrated slope and offset parameters to convert base-10 integers into meaningful physics/engineering variables (such as spacecraft bus uptime clocks, experiments phase indices, and sensor power status).
- **Data Access Isolation Tier (Beacon Data Service Library — .DLL):** The system's explicit Data Access Layer (DAL). It handles connection pooling string definitions, operational query commands, and command transactions directly targeting storage infrastructure. Isolating data infrastructure mechanics to this individual module ensures that underlying database targets can be completely migrated without impacting higher-level orchestration web methods.
- **Persistence Tier (MySQL Relational Database):** The high-volume backend data engine running instance versions optimized for relational storage. The system maintains indexed schema tables that partition raw text payloads, operational tracking coordinates, parsed parameters, lookup dictionaries, and programmatic query logs.

3. Core Functional Feature Requirements

- **Operator Submission Engine:** Ingest hex payload packets and map them alongside operator profile attributes. The system must run automated regular expression validations for validation of 6-digit alphanumeric callsigns and emails, returning a custom QSL confirmation receipt featuring the official mission insignia upon database commit.
- **Decoupled Automated Archival:** The platform must execute standalone parsing and database serialization routines to preserve asynchronous telemetry records securely without degradation during periods of high-frequency data collection.
- **Aggregated Analytical Summaries:** BPS must support automated data reporting queries to calculate platform metrics, tracking global country-of-origin distributions, unique operational callsign totals, and cumulative archived telemetry counts.
- **Telemetry Parameter Mining:** Provide advanced historical filtering tools allowing public researchers to scan archived metadata records filtered by spacecraft ID, timestamp intervals, and discrete telemetry channels.
- **System Diagnostics Logging:** The architecture must run background analytical logging routines to capture distinct session identifiers, query parameters, performance metrics, and application access patterns.

4. Relational Database Schema Infrastructure

The infrastructure relies on indexed relational tables designed to separate incoming payloads, decoded telemetry logs, and tracking parameters:

Table Structure: RawData

Field Name	Data Type	Operational Definition & Constraints
RawSeqID	INTEGER	Unique sequential row identifier (Index 1)
ContributorSeqID	VARCHAR(6)	Foreign key mapping to contributor index
RawData	LONGVARCHAR	The raw, unparsed telemetry hexadecimal block string
RawTimeStamp	DATETIME	System record creation date and time stub

Table Structure: PacketFormat

Field Name	Data Type	Operational Definition & Constraints
PFSatelliteID	VARCHAR(20)	Target spacecraft operational designation key
PFFieldName	VARCHAR(50)	The human-readable engineering variable name
PFSeqID	INTEGER	Unique sequential descriptor field key (Index 1)
PFFirstStartingPosition	INTEGER	The bit/byte offset starting position index
PFFieldLength	INTEGER	Bit/byte width assignment of the data element
PFWellNumber	VARCHAR(4)	Biological well identifier association index
PFFieldType	VARCHAR(20)	System classification descriptor type
PFFieldUnit	VARCHAR(20)	Scientific unit representation (Volt, Kelvin, sec)
PFCalibrationOffset	INTEGER	The calibrated linear equation offset constant

Table Structure: Contributors

Field Name	Data Type	Operational Definition & Constraints
ContributorSeqID	INTEGER	Primary Key identifier for operator profile logs
FullName	VARCHAR(100)	Full name of the submitting terminal operator
CallSign	VARCHAR(6)	Validated 6-digit

EmailAddress	VARCHAR(100)	alphanumeric radio call sign The primary contact email address of the operator
Location	VARCHAR(100)	Country of origin tracking name
Password	VARCHAR(10)	Secure authentication access token

Table Structure: PacketHeader

Field Name	Data Type	Operational Definition & Constraints
PacketSeqID	INTEGER	Unique sequential header layout key (Index 1)
RawSeqID	INTEGER	Foreign key index mapping directly to RawData table
PHWebSite	CHAR(14)	The satellite web mission source classification link
PHBusTime	INTEGER	On-board satellite hardware runtime telemetry clock
PHHealthPowerStatus	INTEGER	Aggregated electronic power system status indicator
PHHealthStartupCounter	INTEGER	Spacecraft flight computer reboot counter log
PHHealthSpacecraftID	INTEGER	Programmatic verification ID of the physical satellite
PHHealthExperimentPhase	INTEGER	Current biological experimental execution sequence phase
PHWellNumber	INTEGER	Target telemetry data well reference number

Table Structure: PacketReadings

Field Name	Data Type	Operational Definition & Constraints
PacketSeqID	INTEGER	Relational sequential identifier link (Index 2)
PRTYPE	VARCHAR(20)	Telemetry parameter metric category type (Index 2)
PRData	INTEGER	The parsed base-10 numerical data parameter value

Table Structure: SubmissionComments

Field Name	Data Type	Operational Definition & Constraints
PacketSeqID	INTEGER	The relational packet index reference linkage
ContributorSeqID	INTEGER	The contributing operator

Comments	LONGVARCHAR	profile relational key Freeform text field logging user notes and signal configurations
-----------------	-------------	--

Table Structure: Codes

Field Name	Data Type	Operational Definition & Constraints
CodeType	VARCHAR(10)	Primary Key data category descriptor group
CodeID	VARCHAR(10)	Primary Key dictionary lookup identification value
CodeDescription	VARCHAR(50)	The expanded human-readable textual translation name

5. Project Administration, Team & Media Archives

The engineering lifecycle, module design, and structural deployment profiles of the Beacon Data Processing System were developed and reviewed by the faculty advisors and student engineering cohorts of Ohlone College in collaboration with associated aerospace laboratories:

Project Faculty Advisors:

- Dr. Jim D. Pham — Ohlone College Computer Science Department Chair / NASA Advisor
- Professor David Topham — Ohlone College Computer Engineering Faculty Advisor

Core Software Engineering Roster:

- Edward Anthonysamy — Software Systems Infrastructure Engineer
- Bruce Davis — Backend Database & Schema Orchestrator
- Gopalakrishnan Veeraragavan (Veeraragavan Gopalakrishnan) — Logic Tier & Parsing Architecture Engineer
- Ryan Edge — Frontend UI & Verification Forms Developer

Historical Photographic & Asset Archive:

The official project files maintain an absolute historical file record under index 'NASA-CREST-2-Project-2009.jpg'. This media asset documents the collaborative engineering cohort and faculty advisors standing side-by-side inside the Ohlone College computer laboratory facilities. The cohort profile is framed underneath the active PharmaSat Spacecraft Project Portal network view (<http://www2.ohlone.edu/people/jpham/crest/>), establishing the historical timeline of the institutional alliance with Santa Clara University and the NASA/CREST aerospace initiative.

6. Specification Archive Note

This technical specification asset represents an immutable historical system record of the 2008-2009 design phase. All software components, database models, class compilation schemas, and configuration metrics detailed herein are preserved for institutional reference.